

Flow-Based Zn – CO₂ Battery for Simultaneous CO₂ Mitigation, Power Generation, and Hydrogen Production

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Abstract— Metal–CO₂ batteries hold significant promise in addressing global warming by enhancing the conversion of CO₂ into value-added products, such as fuels and chemicals. However, conventional Zn – CO₂ battery¹ configurations are limited by their closed-system design, poor scalability, and constraints related to electrolyte concentration and CO₂ purging pressure. To overcome this, we develop a flow-based Zn – CO₂ battery model integrating laminar flow dynamics and electrochemical processes, enabling simultaneous CO₂ mitigation, electricity generation, and hydrogen production. In this system, gaseous CO₂ dissolves in water within a separate tank, forming HCO₃[–], CO₃^{2–}, and H⁺ ions. This lowers the pH and establishes an acidic environment at the cathode, which promotes the hydrogen evolution reaction (HER) and enables sustainable H₂ generation alongside

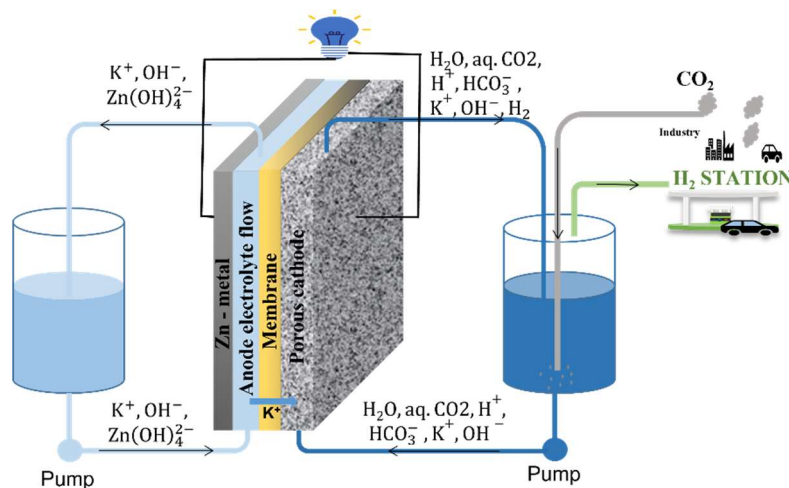


Figure 1: Schematic of Zn-CO₂ flow system

CO₂ utilization. Herein, the system consists of an electrolyte containing a solution of Zn(OH)₄^{2–}, K⁺, and OH[–], where Zn oxidizes to Zn(OH)₄^{2–}, a cation-selective membrane that permits K⁺ ion transport while maintaining ionic separation and a catholyte comprising aqueous CO₂, K⁺, OH[–], H⁺, and HCO₃[–], where HER occurs at the cathode. A schematic is shown in Figure 1. A detailed parametric analysis is performed to optimize battery performance, examining the effects of flow rate, channel volume, applied voltage, and exchange current density. Additional simulations assess species concentration profiles, electrolyte potential, and current distribution, offering insights into system dynamics and performance. This flow-based configuration enhances scalability and provides a practical platform for integrating CO₂ utilization, power, and hydrogen fuel production.

Key words: Zn-CO₂ electrochemical system; Flow batteries; Energy Conversion; CO₂ utilization

Reference:

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